

Iqbal Maqbool Sofi

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Profile

Machine Learning Researcher with an M.S. in Data Science and industry experience in analytics and software engineering. Working at the intersection of robust machine learning, generative models, semantic communication, and 3D representation learning. Experienced in PyTorch-based experimentation, research-code debugging, and controlled model evaluation under severe information loss.

Education

Rowan University Glassboro, NJ *M.S. in Data Science* 2024 – 2026

Amity University India *B.Tech. in Computer Science and Engineering* 2017 – 2021

Research Experience

Semantic Communication & 3D Generative Modeling Rowan University *Machine Learning Research* 2026

- Investigated extension of the Discernment semantic communication framework to ModelNet40 3D point clouds represented by **4,096 discrete MaskedVQVAE3D tokens** per shape.
- Implemented a **MONAI-style decoder-only transformer (DoT)** and benchmarked it against NanoGPT under token loss levels from **50% to 99.2%**.
- Diagnosed a transformer output-head dimensionality mismatch and traced anomalous early classifier performance to an incorrect token dataset, restoring evaluation on the reconciled dataset.
- Demonstrated that despite reaching nearly identical language-model validation loss (~1.07), NanoGPT retained a **6–12.5 percentage-point downstream accuracy advantage** over MONAI under severe corruption.
- Verified token-to-geometry decoding paths and identified an evaluation limitation involving class-sorted 200-sample subsampling versus the **67.9% full-validation clean baseline**.

Selected Projects

Trust-Aware Machine Learning Pipeline Python, Data Reliability

- Analyzed **541,909 retail transaction records** and built a reproducible data-cleaning, feature-engineering, modeling, and evaluation workflow emphasizing reliability, data quality, leakage prevention, and transparent evaluation.

Generative Audio Modeling (AudioMNIST) PyTorch, VQ-VAE, Transformers

- Developed a VQ-VAE and transformer-based audio-generation pipeline using AudioMNIST; evaluated generated samples using a downstream classifier reaching a verified **98.5% accuracy**.

Professional Experience

Accenture India *Application Development Analyst* Feb 2022 – Aug 2024

- Supported analytics, tracking, and data-governance work across more than **80 pharmaceutical websites**.
- Contributed to analytics and tracking implementations for a major UK banking client.
- Collaborated across technical and business teams to ensure data accuracy, quality compliance, and reporting integrity.

DXC Technology India *Associate Professional Software Engineer* Aug 2021 – Feb 2022

- Developed software applications utilizing **C#, .NET, SQL Server, and REST APIs** within an Agile software development framework.

Technical Skills

ML / Research: PyTorch, Deep Learning, Transformers, Vision Transformers, VQ-VAE, Generative Models, Semantic Communication, 3D Representation Learning, Model Evaluation

Programming & Data: Python, SQL, C#, .NET, Pandas, NumPy

Tools & Platforms: Git, Linux, CUDA, Jupyter, Agile, REST APIs